IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2020

MSc and EEE/EIE PART IV: MEng and ACGI

SELECTED TOPICS IN POWER SYSTEMS

Wednesday, 6 May 10:00 am

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer ALL questions.

All questions carry equal marks

Any special instructions for invigilators and information for candidates are on page 1.

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Answer ALL Questions

Question 1 on Flexible AC Transmission System (FACTS)

1. a) A thyristor-controlled series capacitor (TCSC) typically consists of a fixed capacitor in parallel with a thyristor-controlled reactor. Sketch the effective reactance offered by such a TCSC if the thyristor firing angle (α) is varied between 0° and 180°. Label the relevant points and range of interest on the plot.

[2]

b) Explain how a Static Synchronous Series Compensator (SSSC) installed on a transmission line can exercise power oscillation damping (POD). State the POD control law for the SSSC operating in bang-bang mode and support your explanation by sketching the typical dynamic variation of power angle (δ) across the line, the active power flow (P) through it and modulation of the control parameter of the SSSC. Include two traces in each plot showing the cases with and without power oscillation damping control through SSSC.

[3]

c) Draw the phasor diagram for a series compensated line with the line reactance (X) and capacitive series compensation (X_C), both split into two equal halves and assuming the voltages at the sending and receiving ends of the line to be equal in magnitude. Show the voltages at the sending end, receiving end and the midpoint of the line, voltage drops and current on the phasor diagram. Neglect line resistance and shunt charging.

[2]

d) Using the phasor diagram in part (b), derive the expression for the reactive power (Q_C) exchanged with the series compensator in terms of the power angle (δ) across the line, line reactance (X), compensation level (k) and the voltage magnitude (V) at both ends of the line. Neglect line resistance and shunt charging.

[2]

e) A TCSC is installed on a transmission line with reactance X=0.25 p.u and negligible resistance and shunt charging. The voltages at both ends of the line are maintained at 1.0 p.u. Determine the following neglecting line resistance and shunt charging:

i) compensation level (k in %) of the TCSC required to double the maximum active power transfer limit of the line compared to that without the TCSC.

[2]

ii) reactive power exchanged with the TCSC at the point of maximum possible active power transfer.

[2]

iii) critical clearing angle (in degrees) with zero compensation level ($k = 0$) of the TCSC for a power flow of 2.5 p.u. considering the same line reactance in pre- and post-fault conditions and zero power transfer during the fault.

[3]

iv) critical clearing angle (in degrees) with $k = 0.25$ for a power flow of 2.5 p.u. considering the same line reactance in pre- and post-fault conditions and zero power transfer during the fault.

[2]

v) the allowable pre-fault power flow with $k = 0.25$ if the critical clearing angle were to be the same as in part (iii)

[2]

Question 2 on System Operation with Low Inertia

2. a) Battery storage systems can provide frequency response much faster than thermal generators, therefore providing a service called Enhanced Frequency Response (EFR) in the UK. EFR reduces the need for the regular frequency response (R), following this rule：

$$\left(\frac{H}{f_0} - \mathrm{EFR} \cdot \frac{\mathrm{T_{EFR}}}{4 \cdot \Delta f_{\max}}\right) \cdot R \geq \frac{(P_{\mathrm{L}} - \mathrm{EFR})^2 \cdot \mathrm{T_d}}{4 \cdot \Delta f_{\max}}$$

Nomenclature

H: system level of inertia (Units: MW · s).
R: amount of frequency response scheduled (Units: MW).
EFR: amount of EFR scheduled (Units: MW).
P_{L}: value of the generation outage we want to secure against (Units: MW).
f_0: nominal electric frequency of the power grid (Units: Hz).
$\Delta f_{\max}$: maximum admissible frequency deviation to avoid load shedding (Units: Hz).
$\mathrm{T_d}$: Delivery-time of frequency response (Units: s).
$\mathrm{T_{EFR}}$: Delivery-time of EFR (Units: s).

Assumptions:
1. The cost of frequency response (R): £5/MW,
2. The largest generation unit in the UK: 1.8 GW,
3. Other numerical values: f_0 = 50 Hz, Δf_{max} = 0.8 Hz, T_d = 10 s, T_{EFR} = 0.5 s.

i) If the system inertia is 150GWs, i.e. H = 150 GWs, there is a provider to offer 200MW of EFR, what is the maximum bid price (£/MW) that this provider can choose in order to be selected by the system operator? Assume that the aim of the system operator is to minimize the total cost.

[4]

ii) Discuss how the maximum bid price from the provider, found in i), will change if the capacity of the largest generation loss (P_L) increases?

[3]

iii) Assume the system operating condition as in i) and the following:

Wind output	13,000 MW
Demand	25,000 MW
Inflexible generation	
• Must run (e.g. nuclear)	8,400 MW
• Maximum Frequency Response (R)	0 MW
CCGT units	
• Rated output	500 MW
• Minimum Stable Generation	300 MW
• Maximum Frequency Response (R)	100 MW

If the system operator purchased this 200 MW EFR, how much wind curtailment can be reduced?

[5]

b) Answer the following questions related to the implementation of frequency support services in renewable generation systems:

i) A wind turbine with a power rating of 6 MW is controllerd to emulate a synthetic inertia of H=4s. Determine how much the power order would change (ΔP) during a frequency excursion with a RoCoF of -1 Hz/s in a 50 Hz power system.

[3]

ii) Draw the approximate graph of the kinetic energy stored in a wind turbine versus the wind speed. Explain the shape of the curve.

[2]

iii) Consider a wind turbine with an approximate moment of inertia of $I_G = 3.8 \cdot 10^6$ kg·m^{-2} rotating at ω_T=20 min^{-1}. Calculate how much additional power (ΔP constant) the turbine would be able to deliver to support the grid for a period of 2 seconds if the final speed of the turbine was 2% lower

[3]

Question 3 on Power System State Estimation

3. a)

i) What is the purpose of static state estimation in power systems?

[4]

ii) What quantities are estimated in static state estimation?

[3]

iii) List the major steps in static state estimation.

[3]

b)

i) List three important challenges in distribution system state estimation.

[3]

ii) List three main statistical criteria for assessing the effectiveness of an algorithm in distribution system state estimation.

[3]

iii) Explain one of the statistical criteria for assessing the effectiveness of a state estimation algorithm in power distribution context.

[4]

Question 4 on Integrated Energy Systems

4. **Part I**

a) List and briefly describe the key challenges associated with integration of wind generation in the future UK system, considering both short term operation and long term planning time horizons.

[4]

b) What are the advantages and disadvantages of thermal energy storage compared to Electricity Storage?

[3]

Part II

Considering the electrification of heat sector, heat systems could provide flexibility to the electricity system and support more cost-effective integration of variable wind generation, through thermal energy storage (TES). Consider a system with capacity of wind generation of 35GW, nuclear generation 12.4GW, with the rest of the load being supplied by a number of CCGT plantsm, which are all the same type with same characteristics. Electricity demand includes non-heat demand and electrified heat demand. Operating reserve is required to deal with uncertainties in wind generation production, which is managed by part-loading CCGT.

Key characteristics of the CCGT plant are:

Capacity: 550 MW
Minimum Stable Generation (MSG): 300 MW
Start-up time: 1 hour (it takes 1 hour for a CCGT plant to start-up)

System operational conditions and reserve requirements are given in the table below, considering 2 consecutive hours of system operation.

	1st hour	**2nd hour**
Wind available (MW)	28,800	14,400
Nuclear (MW)	12,400	12,400
Electricity Demand (MW) - Non-heat demand - Electrified heat demand	51,000 36,000 15,000	51,000 36,000 15,000
Reserve requirement (MW)	14,400	9,600

a) Assuming that TES is not available, answer the following questions:

i) How many CCGT plants must be online in the 2nd hour?

[3]

ii) How much wind would be curtailed in the 1st hour (consider CCGT start-up time)?

[3]

b) Consider that TES is available. The efficiency of TES is 95%, i.e., 5% of stored thermal energy is lost per hour.

i) Calculate the minimum (power equivalent) capacity of TES that would be needed to ensure no wind is curtailed in the 1st hour.

[3]

ii) How many CCGTs have to be online at the 2nd hour?

[2]

iii) If CO_2 emission associated with the operation of CCGT is 0.4ton/MWh, how much CO_2 is saved by using TES?

[2]